

Ocean Networks Canada: Holyrood Acoustic Testing Summary

Executive summary

At the Holyrood Bay platform, contamination of hydrophone audio data by noise from other instruments was observed and reported in January 2024 in Jira ticket [INSTR-2436](#). This included both narrowband tonal and broadband noise sources shown in the following spectrograms. The sound level and frequency content of these noise sources is sufficient to mask other signals of scientific interest such as natural ambient sound, biological vocalizations, and anthropogenic sources. The suspected noise sources suggested in [INSTR-2436](#) included the ADCP, pCO₂ sensor, and Seabird Microcat, though the exact sources were unknown at the time of reporting.

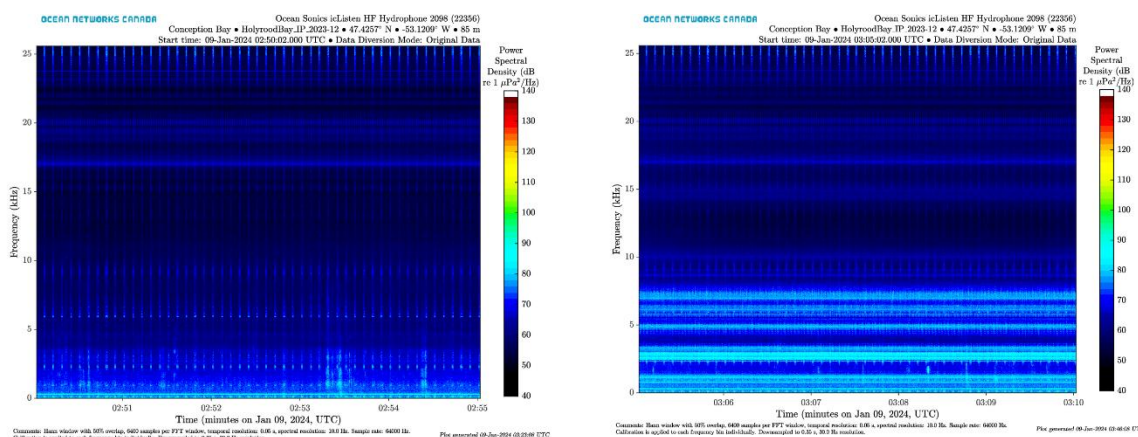


Figure 1: Spectrograms of noise contamination of hydrophone data at the Holyrood Bay platform. (Left) Constant signal. (Right) Intermittent signal every 2.5 hours.

To determine the source(s) of this noise, testing was undertaken to power on each instrument independently and record each instrument's acoustic signature with the hydrophone (see Jira ticket [SYS-17399](#)). Each instrument was then powered down before moving onto the next instrument for testing, to ensure each instrument was measured in isolation. Table 1 on the following page summarizes the findings in order of level of concern with respect to hydrophone audio data quality.

Recommendations for instruments with 'Medium' and 'High' levels of concern:

- Investigate modifications to instrument configurations to reduce noise levels.
- Discuss modifications of sampling frequency and/or duty cycle with the relevant principal investigators to enable more 'noise-free' time for hydrophone recording.

Table 1: Summary of Holyrood acoustic test findings

Device	Device ID(s)	Noise detected?	Noise description	Frequency range	Level increase above ambient	Level of concern
Seabird Microcat	13116	Yes	Continuous tonal whirring noise with broadband components, ~6s interval.	20 Hz – 32 kHz	Up to 40 dB	High
pCO2	71960	Yes	Broadband whirring noise for ~5-minutes to 1-hour+, with intermittent quiet periods.	350 Hz – 32 kHz	Up to 30 dB	High
AML1-RT + Fluorometer	50500, 47360	Yes	Continuous broadband high-frequency (HF) noise with narrowband components.	16 kHz+	6 – 20 dB	Medium
ADCP	47140	Yes	Continuous broadband HF clicking, ~0.5s interval.	6 kHz+	~20 dB	Medium
SubC Imaging Rayfin + Lights	62300, 84172	Yes	Initially there was no noise associated with this device. However, a low-frequency tonal noise built-up over time and stopped abruptly when the device was powered down.	293 Hz	Up to 15 dB	Medium
JB 031	-	Yes	Continuous narrowband HF tonal which is modulated depending on power configuration.	~23 kHz	25 dB	Low
Teledyne Benthos modem	50480	Yes	Series of loud tones on startup, quiet otherwise.	9 – 14 kHz	90 dB+	Low
Seabird SeapHOX	73700	No	-	-	-	None

1. Introduction

At the Holyrood Bay platform, contamination of hydrophone audio data by noise from other instruments has been observed and initially reported by Herminio Foloni Neto in January 2024. To determine the source of this noise, a test procedure was developed by Herminio (later modified by Brendan) to power cycle each instrument independently while continuously recording with the hydrophone. The test was performed on July 21-22, 2025. This procedure allowed the acoustic signature of each device to be measured. The methods and results of this test are presented in this document. A discussion follows to suggest next steps to improve hydrophone audio data quality.

2. Methods

The following test procedure was performed by Patrick Frisby on the Observatory Systems team with audio analysis during the test by Brendan Smith on the Data Analytics and Quality team. Time stamps for each step are listed. Audio files and spectrograms corresponding to each step can be found here: [Holyrood acoustic testing audio files](#).

Table 2: Acoustic testing procedure

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Step ID	Description	Device ID(s)	Time start (UTC)	Time end (UTC)	Date (UTC)
1	Turn off all instruments except the hydrophone	22356	16:26	17:32	21-Jul-2025
2	Wait for 1 hour				
3	Turn on the ADCP	47140	17:32	18:11	
4	Wait for 30 min	47140			
5	Turn off the ADCP	47140	18:11	18:17	
6	Wait for 5 min				
7	Turn on the Teledyne Benthos modem	50480	18:17	18:49	
8	Wait for 30 min	50480			
9	Turn off the Teledyne Benthos modem	50480	18:49	18:55	
10	Wait for 5 min				
11	Turn on the Seabird Microcat	13116	18:55	19:27	
12	Wait for 30 min	13116			
13	Turn off the Seabird Microcat	13116	19:27	19:34	
14	Wait for 5 min				
15	Turn on Seabird SeapHOX	73700	19:34	20:06	
16	Wait for 30 min	73700			
17	Turn off Seabird SeapHOX	73700	20:06	20:14	
18	Wait for 5 min				
19	Turn on the pCO2	71960	20:14	20:47	
20	Wait for 30 min	71960			
21	Turn off the pCO2	71960	20:47	20:56	
22	Wait for 5 min				
23	Turn on all devices listed in steps 3-21	See above	20:56	17:07 (+1)	21/22-Jul-2025
24	Wait 4 hours+				
25	Turn off all instruments except the hydrophone	22356	17:07	17:42	22-Jul-2025
26	Wait for 30 minutes (all other instruments should be off, but JB 31 should still have power - this is to capture the natural ambient background noise today, which may have changed from July 21)				

27	Power off JB 031 entirely (still no instruments on except hydrophone)		17:42:13	18:26:52	
28	Wait for 30 minutes (leave JB 31 off for remainder of this test if possible)				
29	Turn on AML1-RT + Fluorometer	50500, 47360	18:26:52	19:02:19	
30	Wait for 30 minutes	50500, 47360			
31	Turn off AML1-RT + Fluorometer	50500, 47360	19:02:19	19:09:30	
32	Wait for 5 minutes				
33	Turn on SubC Imaging Rayfin + Lights	62300, 84172	19:09:30	19:40:40	
34	Wait for 30 minutes	62300, 84172			
35	Turn off SubC Imaging Rayfin + Lights	62300, 84172	19:40:40	19:47:19	
36	Wait for 5 minutes				
37	Power on JB 031		19:47:19	19:58:39	
38	Wait for 10 minutes				
39	Power on all remaining instruments		19:58:39	—	

3. Results

Time series and spectrograms for the primary steps during the testing procedure are shown below. Note: all audio files displayed below have been normalized to -3 dB peak amplitude for visualization purposes. This means relative loudness comparisons between different recordings are not possible from these visualizations.

3.1. Ambient day 1

On day 1 (July 21, 2025), there were ships in the vicinity of the hydrophone which were audible in the recordings and visible on spectrograms. They moved past the hydrophone at varying speeds, which affected their received loudness at the observation platform over time. In between ship presence, there were relatively quiet periods with distant ship noise.

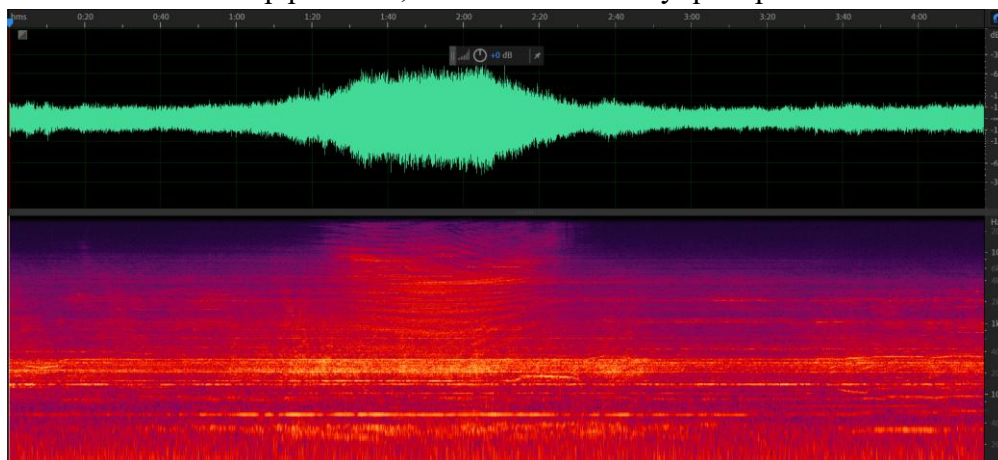


Figure 2: Ambient noise day 1 - ship example (time series (top), spectrogram (bottom)).

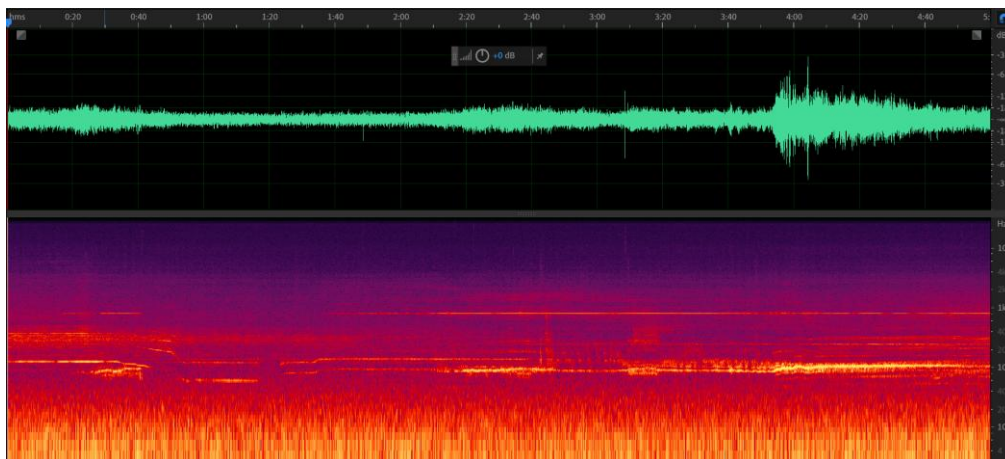


Figure 3: Ambient noise day 1 - relatively quiet periods between ship noise (time series (top), spectrogram (bottom)).

3.2. ADCP

The ADCP is detected by the hydrophone. Its acoustic signature is characterized by a broadband high-frequency click with a ~ 0.5 s interval between clicks. The dominant frequency of this signal is approximately 20 kHz. There is acoustic energy between ~ 6 kHz up to the Nyquist frequency of 32 kHz.

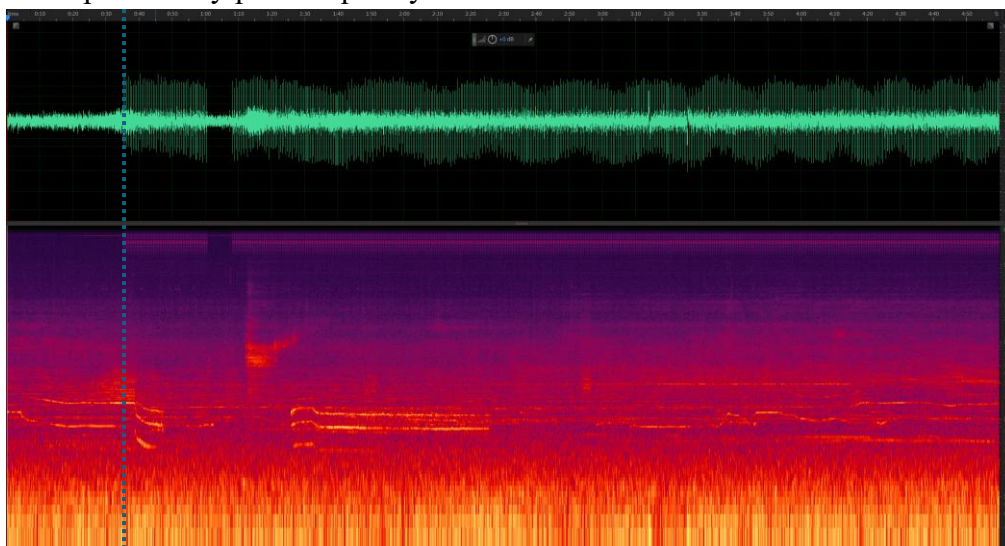


Figure 4: ADCP acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.3. Teledyne Benthos modem

The acoustic modem generates four loud narrowband tones when it is first powered on, but is quiet otherwise. Presumably, the device was not transmitting any data while it was on. If it was transmitting data it would likely generate similar loud tonal signals.

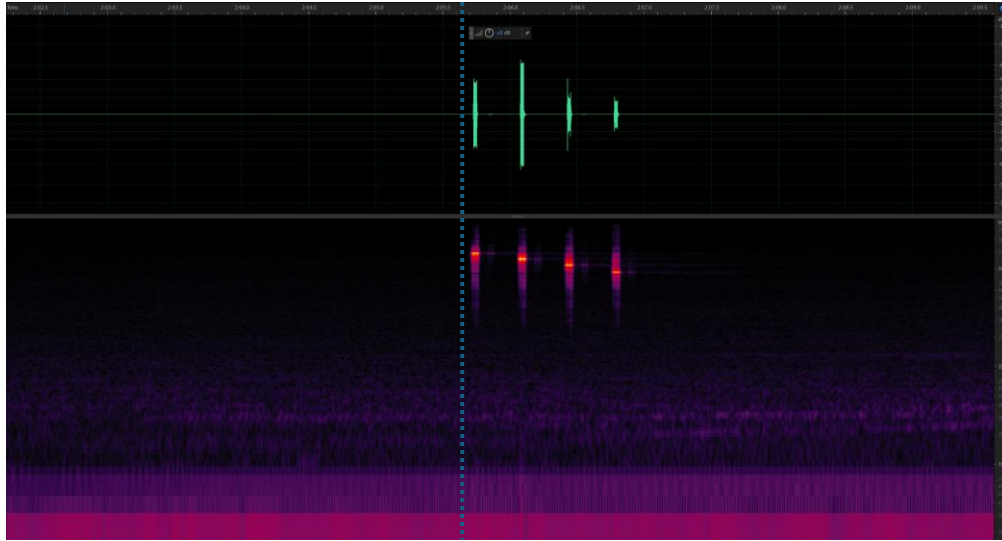


Figure 5: Modem acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.4. Seabird Microcat

The Seabird Microcat generates a loud tonal whirring noise with broadband components. The noise occurs every 6 seconds continuously during the operation of the device. This noise is likely produced by a pump on the device which operates every 6 seconds.

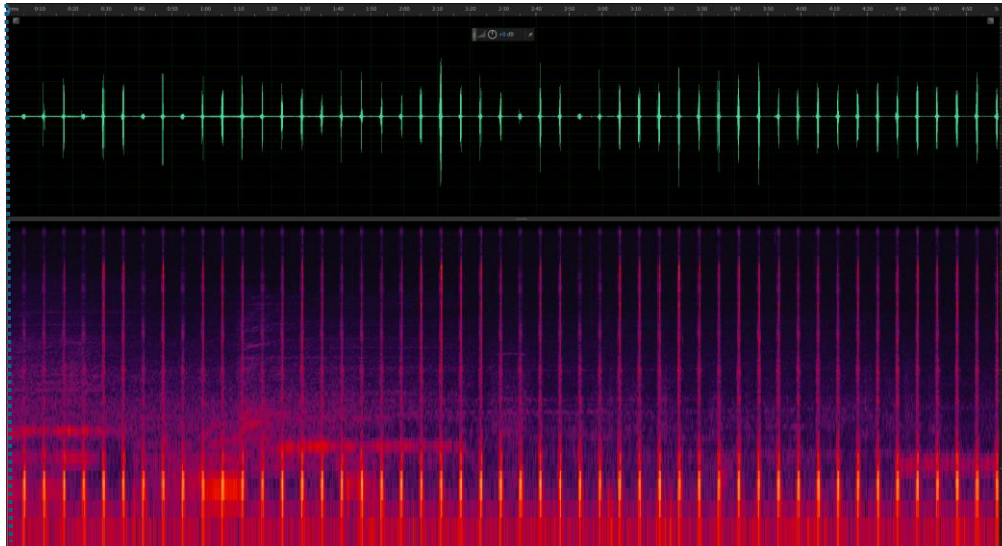


Figure 6: Microcat acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.5. Seabird SeapHOX

No noise was observed associated with the Seabird SeapHOX device. The tonal noise near 100 Hz shown in Figure 7 is due to ship noise.

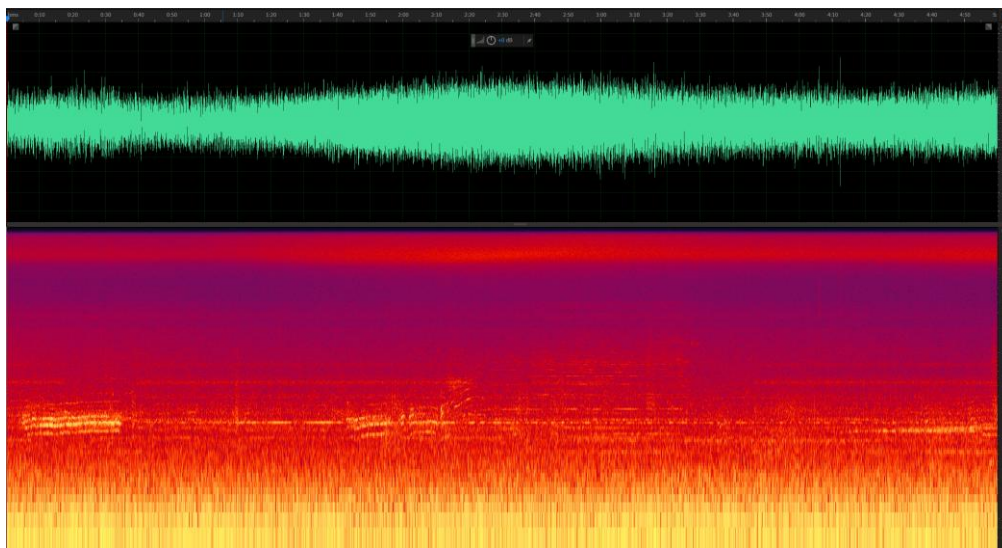


Figure 7: Seabird SeapHOX acoustic signature (time series (top), spectrogram (bottom).)

3.6. pCO₂

During the pCO₂ individual instrument test, a broadband whirring noise was observed which lasted for approximately 5 minutes when first turned on. This signal then ended and did not occur again during the remaining of the 30-minute test. However, at other times (for example during step 39 when all instruments were powered on), the pCO₂ noise continued for more than an hour before stopping.

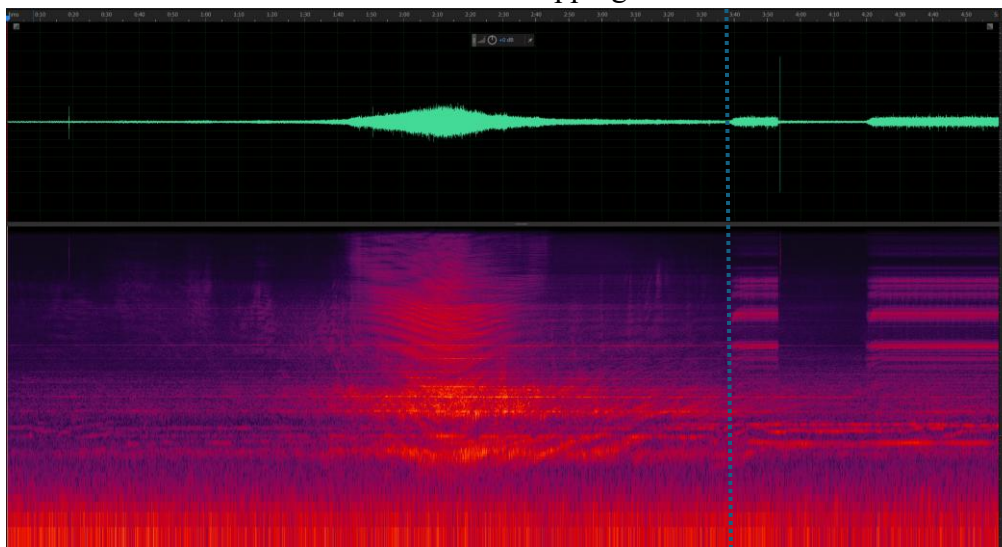


Figure 8: pCO₂ acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.7. All devices on

With all devices on, the signatures of the equipment identified above are audible and visible together in the spectrogram. There is intermittent ship noise also visible in this spectrogram. The dominant noise sources are the Seabird Microcat and pCO₂ sensors, with the ADCP clicks faintly visible above 10 kHz.

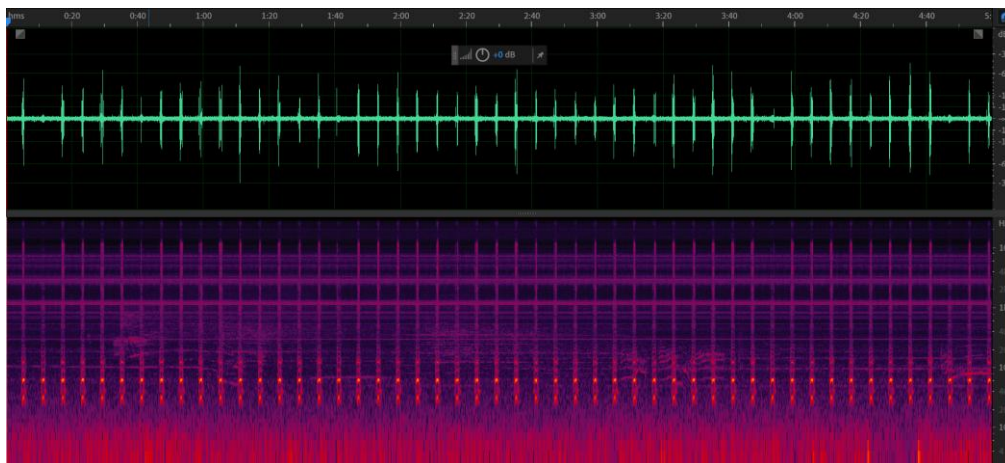


Figure 9: All devices on, day 1 (time series (top), spectrogram (bottom)).

3.8. Ambient day 2

On day 2 (July 22, 2025), there was less ship traffic in Holyrood Bay, resulting in less ship noise received by the hydrophone. Natural sounds were more pronounced on day 2, including the sound of rocks moving on the seabed, likely due to wave action on the shoreline. This signal is shown as intermittent broadband energy in the spectrogram between 1 – 10 kHz. Later in the day, there was also some evidence of rain noise at higher frequencies.

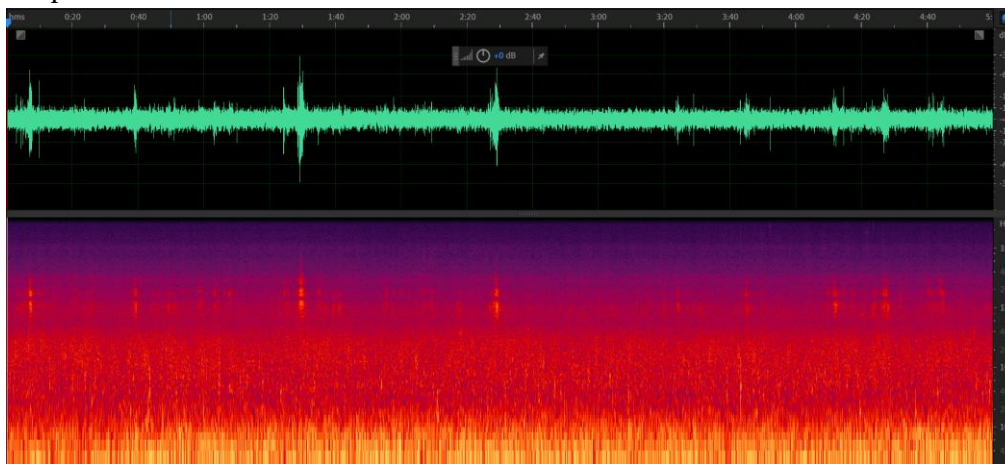


Figure 10: Ambient noise day 2 - quiet with rocks occasionally moving on the seabed (time series (top), spectrogram (bottom)).

3.9. Ambient with JB 031 off

Powering down mini junction box 031 resulted in a change of a high frequency tone in the hydrophone signal. When JB 031 was on, there was a narrowband tone present in the signal at ~ 23.714 kHz. After the junction box is powered down, this tone shifts to ~ 23.290 kHz and is reduced in amplitude by ~ 8 dB. Since this narrowband signal is modulated by JB 031 being powered down, it is likely electrically generated noise associated with the junction box. The continued presence of this noise at a different frequency suggests

electrical noise remains from JB 023 as well, and is affected by the power configuration at the site. Note that the frequency axis of the spectrogram below begins above 22 kHz.

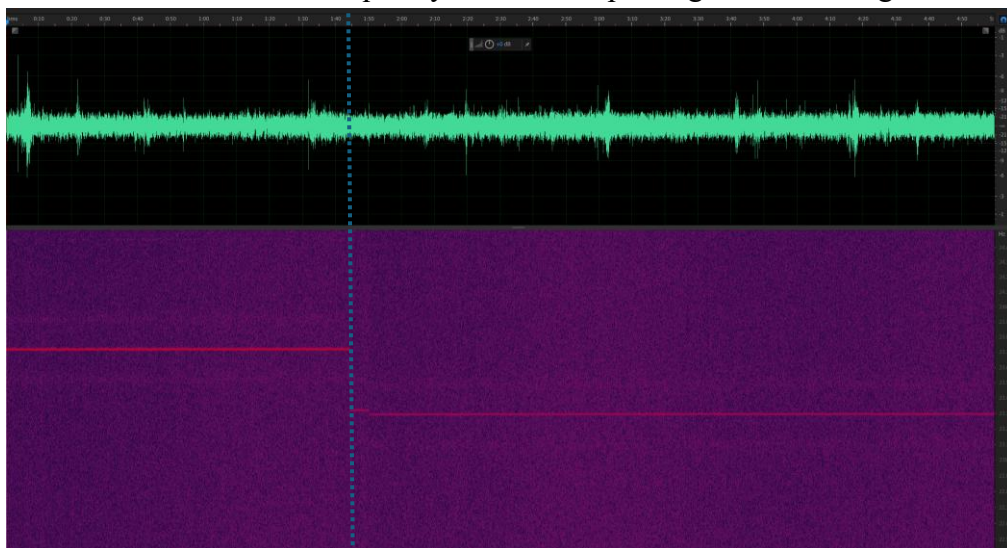


Figure 11: JB 031 acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.10. AML1-RT + Fluorometer

The AML1-RT + Fluorometer generates broadband high-frequency noise above 16 kHz with some additional narrowband tonal components. The noise persisted continuously while the device was operating.

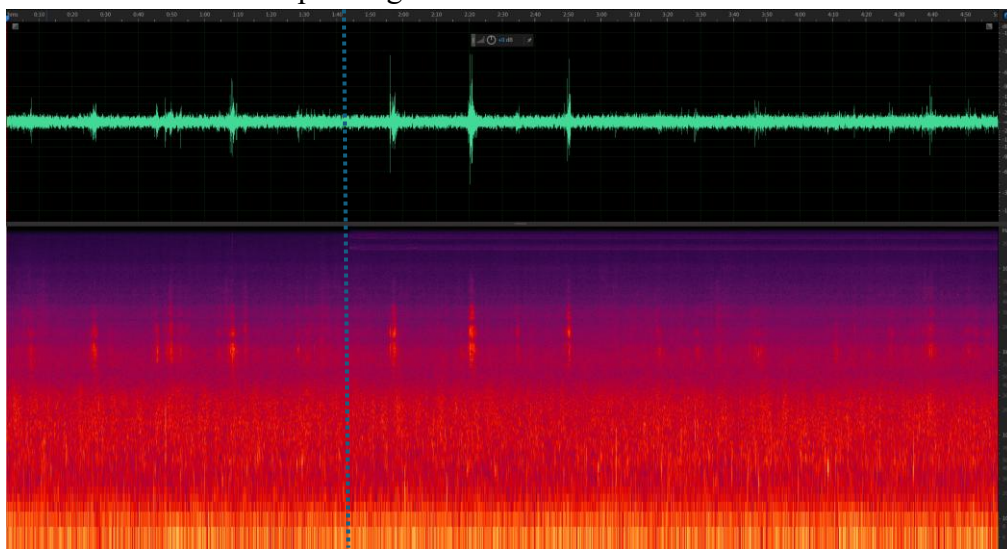


Figure 12: AML1-RT + Fluorometer acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

To better visualize this high-frequency noise, the same plot is shown below with the frequency axis limits adjusted to begin at 14 kHz.



Figure 13: AML1-RT + Fluorometer acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.11. SubC Imaging Rayfin + Lights

The SubC Imaging Rayfin + Lights did not produce any detectable noise when first powered on. However, throughout the 30-minute test period a tonal noise at approximately 293 Hz developed which ended abruptly when the device was powered down. The power on phase is shown in Figure 14.

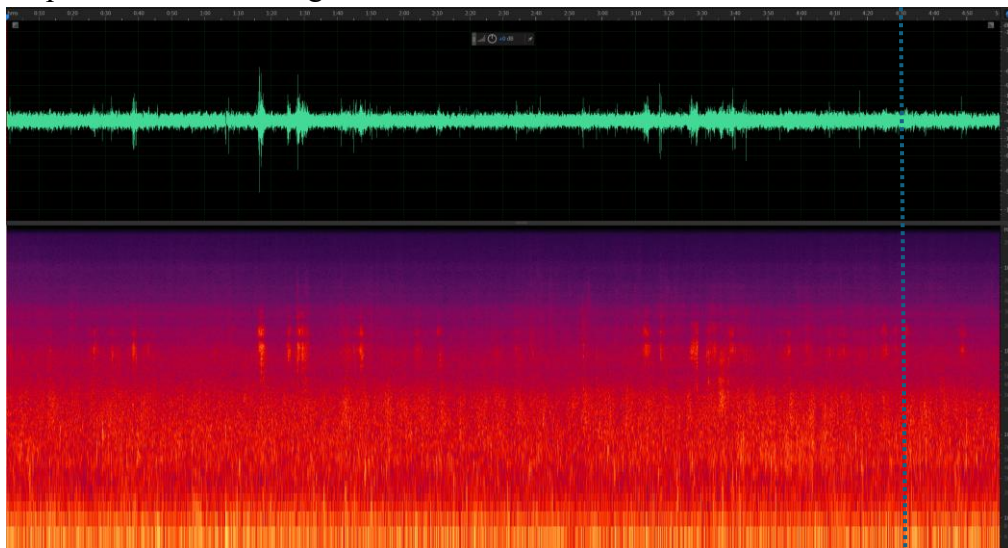


Figure 14: Power on of SubC Imaging Rayfin + Lights acoustic signature (time series (top), spectrogram (bottom). Power on time = blue dashed line.)

The power off phase is shown in Figure 15.

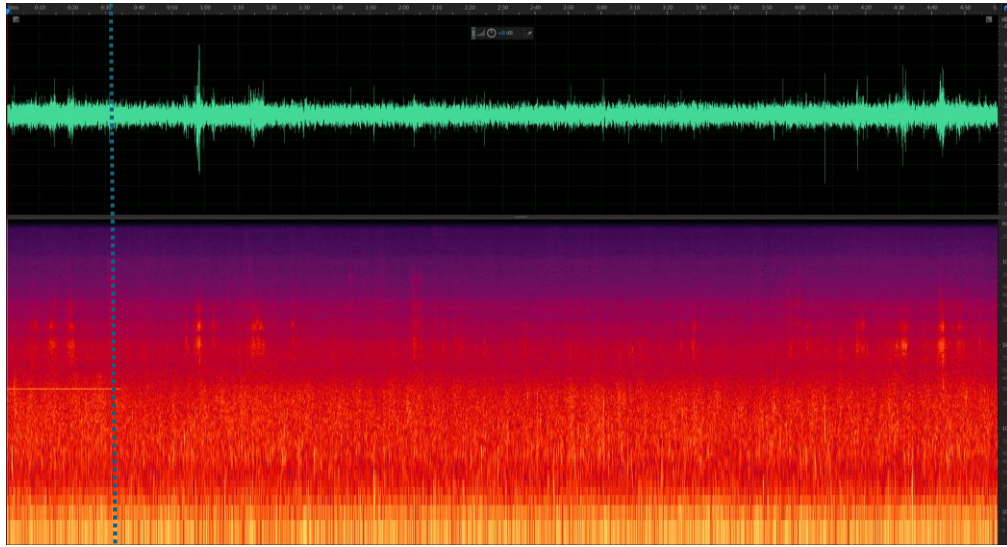


Figure 15: Power off of SubC Imaging Rayfin + Lights acoustic signature (time series (top), spectrogram (bottom). Power off time = blue dashed line.)

3.12. JB 031

When mini junction box 031 is powered back on, the high frequency narrowband tone observed earlier increases in frequency again from ~ 23.290 kHz back to its original state of ~ 23.714 kHz.

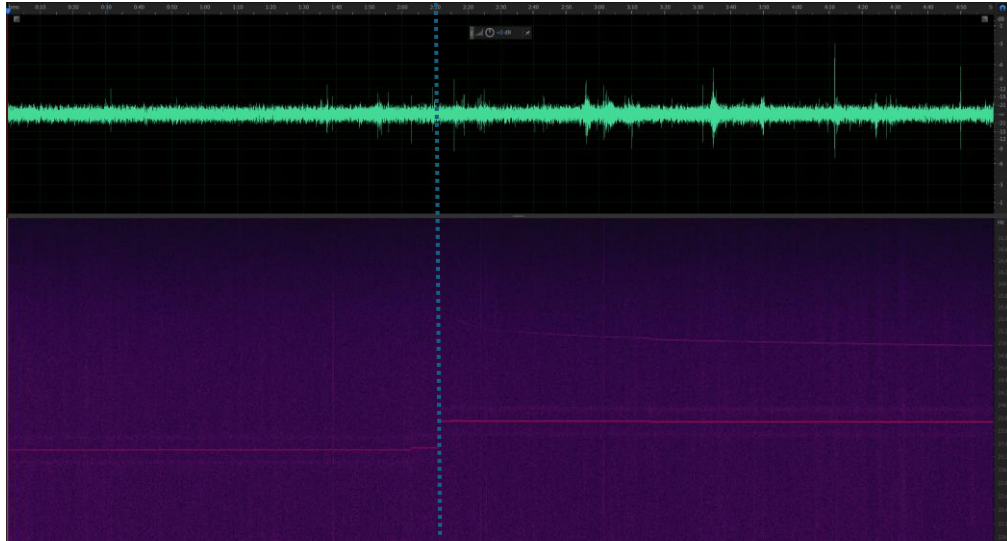


Figure 16: JB 031 acoustic signature (time series (top), spectrogram (bottom). Start time = blue dashed line.)

3.13. All devices on

With all devices on, the noise sources identified above were again visible in the spectrogram together.

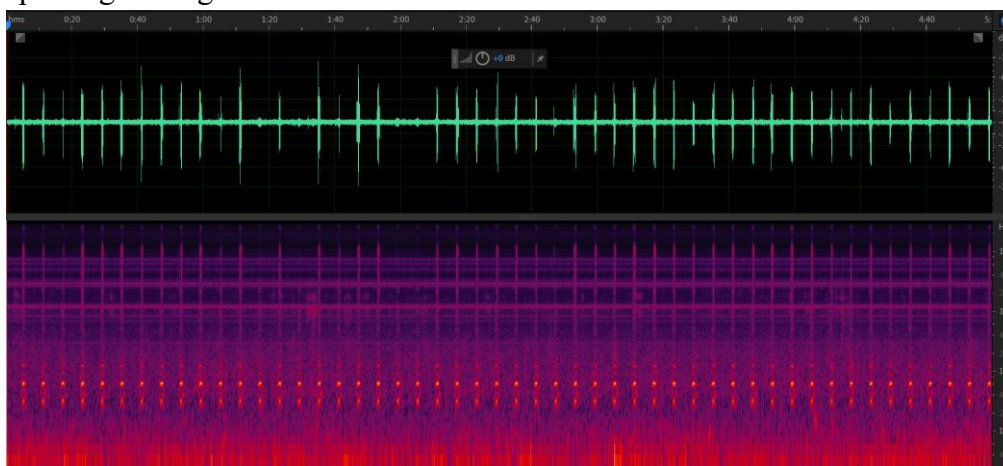


Figure 17: All devices on, day 2 (time series (top), spectrogram (bottom)).

3.14. Summary

The following diagram identifies the primary noise sources in the spectrogram when all are operating together.

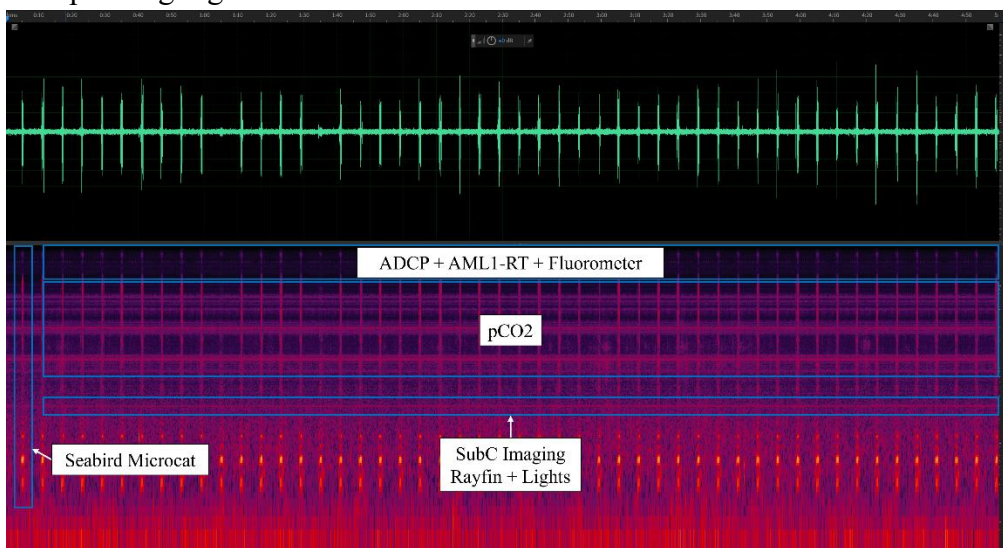


Figure 18: Primary device noise sources summary diagram (time series (top), spectrogram (bottom)).

4. Discussion

The primary acoustic noise sources observed during the testing were the Seabird Microcat, pCO₂, AML1-RT + Fluorometer, ADCP, and SubC Imaging Rayfin + Lights (see Table 1 for details and associated level of concern for each noise source). It is recommended that these noise sources are mitigated either by modifying the instrument configuration into a mode which operates with less noise, by introducing longer sampling intervals, and/or by operating with a duty cycle to allow some ‘noise-free’ time for hydrophone audio recording.

The Seabird Microcat and pCO₂ were assigned a ‘High’ level of concern in the table because their frequency content and amplitude is sufficient to cause the hydrophone audio data to be mostly unusable while they are operating. They both operate for extended periods at times and mask most natural, biological, or anthropogenic acoustic signals which would otherwise be detected by the hydrophone.

The AML1-RT + Fluorometer, ADCP, and SubC Imaging Rayfin + Lights were assigned a ‘Medium’ level of concern because while they may interfere with acoustic data quality, they are generally of lower amplitude and limited to specific frequency ranges, meaning that other frequency bands in the hydrophone data are still usable. However, it is still recommended that noise from these devices is mitigated to ensure that hydrophone audio data is usable for the broadest possible group of users.

Mini junction box JB 031 and the Teledyne Benthos modem were assigned a ‘low’ level of concern because the signals they generate are very narrowband (in the case of JB 031) or infrequent (in the case of the modem). If a user was specifically interested in the frequency band at which JB 031 emits noise, this could be an issue, however this is unlikely to be a concern to most users. Likewise, when the Teledyne Benthos modem operates it is very loud, but during this testing that noise was only detected for a few seconds when it was first powered on. If the modem starts being used more frequently to transmit data, this would likely become a significant noise source in the hydrophone audio data.

5. Conclusions

This acoustic testing protocol of power cycling each instrument independently allowed the noise sources at the Holyrood platform to be identified and quantified. Table 1 summarizes the results of this test. It is recommended that noise produced by instruments with ‘Medium’ and ‘High’ levels of concern is mitigated to improve hydrophone audio data quality.

In addition to identifying acoustic noise sources at this site, this testing has also provided a library of ONC instrument sounds, which has been discussed by both the Science and DAQ teams as a useful library for hydrophone data users. This protocol could be used at other sites to increase our understanding of the noise produced by ONC’s instruments, and to assist hydrophone data users with acoustic source identification.